SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

Technology Stack

Date	29 June 2026
Team ID	xxxx
Project Name	Opticrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack

The chosen technology stack provides a complete environment for agricultural data preprocessing, machine learning model development, web application development, deployment, and project management, ensuring the efficient implementation of the OptiCrop system. It enables accurate crop prediction based on soil andenvironmental parameters while providing a user-friendly interface for farmer

Technology Category	Technology Used	Purpose
Programming Language	Python	Core programming language used for application development and machine learning.
Development Environment	Jupyter Notebook	Model development, experimentation, and testing.
IDE	Visual Studio Code / PyCharm	Writing, debugging, and maintaining source code.
Data Manipulation	Pandas	Data cleaning, preprocessing, and transformation.
Numerical Computing	NumPy	Efficient numerical operations and array processing.
Data Visualization	Matplotlib	Creating graphs and visual representations of data.
Data Visualization	Seaborn	Statistical data visualization and exploratory data analysis.
Machine Learning Library	Scikit-learn	Building, training, and evaluating machine learning models.
Web Framework	Flask	Developing the web application interface for prediction.

Technology Category	Technology Used	Purpose
Version Control	Git	Managing project versions and source code history.
Repository Hosting	GitHub	Team collaboration and project management.
Cloud Platform	IBM Cloud	Hosting cloud services for the application.
Model Deployment	IBM Watson Machine Learning	Deploying the trained model and generating prediction results.
Dataset	Credit Card Application Dataset	Historical applicant data used for trainin and testing the model.
Operating System	Windows 10/11	Development and execution environmen

Machine Learning Algorithms Used

Algorithm	Purpose
Logistic Regression	Baseline classification model for predicting credit card approval.
Decision Tree Classifier	Rule-based classification for approval prediction.
Random Forest Classifier	Ensemble learning algorithm for improving prediction accuracy.
XGBoost Classifier	High-performance boosting algorithm for optimized classification results.